AI Intellicore CMP

Intellicore CMP — GCP Architecture & Cost

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Intellicore CMP — GCP Architecture & Cost to Operate

> Pricing below uses GCP **list prices** for the **asia-south1 (Mumbai)** region as a planning estimate. Verify exact figures with the GCP Pricing Calculator before committing. LLM/AI usage is **not** a platform cost — customers Bring Their Own Key (Settings → AI Keys), so inference is billed to the customer's own Anthropic/OpenAI/Google account.

Two deployment topologies

Intellicore CMP ships in two shapes. Every new customer starts on **Topology A** (cheap, single-VM, isolated) and graduates to **Topology B** (HA managed services) when their scale or SLA warrants it.

Topology A — Single-VM (current default, per-customer isolation)

Everything runs as Docker Compose on one VM behind Caddy (auto-TLS via sslip.io). This is what `deploy-customer.yml` provisions today.

```
flowchart TB
    User([Customer users]) -->|HTTPS| Caddy
    subgraph VM["1× Compute Engine VM (e2-standard-4)"]
        Caddy[Caddy<br/>reverse proxy + TLS]
        Caddy --> FE[Next.js frontend :3000]
        Caddy -->|/api/*| BE[FastAPI backend :8000]
        BE --> PG[(PostgreSQL)]
        BE --> NEO[(Neo4j)]
        BE --> REDIS[(Redis)]
    end
    BE -.->|Bring Your Own Key| LLM[[Anthropic / OpenAI / Gemini<br/>billed to customer]]
    BE -.->|read-only| CLOUD[[Customer GCP / AWS APIs]]
```

Characteristics - One VM per customer → hard data isolation, simplest billing, easiest to reason about. - All state (Postgres, Neo4j, Redis) lives on the VM's persistent disks. - Backups via disk snapshots. - Suitable for Foundation / Advanced tiers and pilots.

Topology B — Production HA (managed services)

For Elite-tier / larger customers needing high availability and horizontal scale.

```
flowchart TB
    User([Customer users]) -->|HTTPS| LB[Cloud HTTPS Load Balancer]
    LB --> FE[Frontend<br/>Cloud Run / GKE]
    LB -->|/api/*| BE[Backend<br/>Cloud Run / GKE]
    BE --> PG[(Cloud SQL for PostgreSQL<br/>Regional HA)]
    BE --> REDIS[(Memorystore for Redis<br/>Standard HA)]
    BE --> NEO[(Neo4j<br/>Aura or self-managed VM)]
    BE -.->|Bring Your Own Key| LLM[[Anthropic / OpenAI / Gemini]]
    BE -.->|read-only| CLOUD[[Customer GCP / AWS APIs]]
    subgraph Obs[Operations]
        MON[Cloud Monitoring + Logging]
    end
    BE --> MON
```

Characteristics - Stateless app tier scales horizontally (Cloud Run or GKE Autopilot). - Managed Postgres & Redis with automated failover, backups, patching. - Neo4j on Aura (managed) or a dedicated VM. - Global HTTPS LB with managed certs + Cloud Armor option.

Component responsibilities

Component	Role
Caddy / Cloud LB	TLS termination, reverse proxy, edge auth gate (session check)
Next.js frontend	UI for all 10 pages; talks to backend via `/api/*`
FastAPI backend	Auth (password + TOTP), Memory graph API, events ingest, settings/BYOK
PostgreSQL	Users, tenants, sessions, cloud accounts, guardrails, **encrypted AI keys**
Neo4j	The operational Memory knowledge graph (events, patterns, causal chains)
Redis	Sessions and caching
BYOK LLM	Customer's own Anthropic/OpenAI/Gemini key powers all AI features

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Network & security

- **TLS everywhere** — Caddy (Topology A) or Google-managed certs (Topology B).
- **Edge auth gate** — every route requires a valid session (forward-auth to `/auth/session/check`); only the login shell and auth endpoints are public.
- **2FA** — email/password + TOTP (Google Authenticator compatible).
- **Per-tenant isolation** — Topology A: separate VM per customer. Topology B: tenant_id scoping on every query + separate managed instances per customer where required.
- **AI keys encrypted at rest** — `pgp_sym_encrypt` (pgcrypto); only last 4 chars ever returned to the UI.
- **Firewall** — only 80/443 open to the internet; databases never exposed publicly.
- **Least-privilege cloud access** — read-only service accounts to poll customer GCP/AWS.

Cost — Topology A (single-VM, per customer)

Monthly, asia-south1 list prices. This is the recurring cost **per customer**.

Line item	Spec	On-demand / mo	With 1-yr CUD / mo
Compute Engine VM	e2-standard-4 (4 vCPU, 16 GB)	~$110	~$70
Boot + data disk	100 GB pd-balanced	~$11	~$11
Static external IP	1 address, in use	~$3	~$3
Disk snapshots (backup)	~100 GB, daily retention	~$6	~$6
Network egress	~50–100 GB/mo	~$10	~$10
Cloud Logging/Monitoring	modest ingest	~$5	~$5
Total (infrastructure)		**~$145/mo**	**~$105/mo**
LLM / AI inference	Bring Your Own Key	**$0 to platform**	**$0 to platform**

≈ $105–145 / month per customer (~₹9,000–12,500/mo). Annual: **~$1,300–1,750**.

A lighter **e2-standard-2** (2 vCPU, 8 GB) works for small/pilot tenants and drops the VM line to ~$55/mo on-demand (~$35 CUD) → **total ~$85–95/mo**.

Cost — Topology B (production HA)

Monthly, asia-south1 list prices, per customer. Ranges reflect HA vs. non-HA choices.

Line item	Spec	Est. / mo
App tier (frontend + backend)	Cloud Run (min instances) or GKE Autopilot	$80 – $160
Cloud SQL for PostgreSQL	db-custom-2-8, regional HA + storage	$250 – $350
Memorystore for Redis	Standard HA, 1–2 GB	$50 – $80
Neo4j	Aura Professional **or** e2-standard-2 self-managed	$55 – $130
Cloud HTTPS Load Balancer	forwarding rules + static IP	$20 – $25
Network egress	100–300 GB/mo	$15 – $40
Cloud Logging/Monitoring	higher ingest	$20 – $40
Backups / snapshots	automated	$15 – $25
Total (infrastructure)		**~$505 – $850/mo**
LLM / AI inference	Bring Your Own Key	**$0 to platform**

≈ $505–850 / month per customer (~₹43,000–73,000/mo). Annual: **~$6,000–10,200**. Committed-use / sustained-use discounts on Cloud SQL and compute can cut 20–37%.

Per-customer economics (Topology A)

Because each customer is one isolated VM, cost scales close to linearly:

Customers	Infra / mo (on-demand)	Infra / mo (CUD)	Annual (CUD)
1	~$145	~$105	~$1,260
5	~$725	~$525	~$6,300
10	~$1,450	~$1,050	~$12,600
25	~$3,625	~$2,625	~$31,500

Shared platform overhead (CI/CD runners, a management project, artifact registry, DNS) adds a flat **~$20–50/mo** regardless of customer count.

> AI inference never appears in these tables — it is always the customer's own spend on their own key.

Recommendations

1. **Default new customers to Topology A** via the `deploy-customer.yml` pipeline. It's cheap (~$105–145/mo), fully isolated, and provisions in minutes.
2. **Buy 1-year committed-use discounts** on steady-state VMs to save ~35%.
3. **Right-size**: pilots and small tenants on `e2-standard-2`; standard on `e2-standard-4`; only move to Topology B when HA/scale is contractually required.
4. **Graduate to Topology B** per customer, not globally — keep the cheap default.
5. **Keep BYOK** — it removes the platform's largest variable cost (LLM tokens) and lets customers control their own AI spend and data-processing agreements.
